Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If A, B, C are three events such that $P(B)=\frac{4}{5}$, $P(A \cap B \cap C^c)=\frac{1}{4}$ and $P(A^c \cap B \cap C^c)=\frac{1}{3}$, then $P(B \cap C)$ is equal to

(A) $\frac{11}{60}$

(B) $\frac{1}{5}$

(C) $\frac{13}{60}$

(D) $\frac{1}{4}$

Q.2

Let p denote the probability that a man aged x years will die in a year. The probability that out of n men $\mathrm{A}_1, \mathrm{A}_2, \mathrm{A}_3, \ldots \ldots \mathrm{A}_{\mathrm{n}}$ each aged x, A_1 will die in a year and will be the first to die is

(A) $\frac{1}{\mathrm{n}}\left[1-(1-\mathrm{p})^{\mathrm{n}}\right]$

(B) $\left[1-(1-p)^{\mathrm{n}}\right]$

(C) $\frac{1}{\mathrm{n}-1}\left[1-(1-\mathrm{p})^{\mathrm{n}}\right]$

(D) none of these

Q.3

Two integers x and y are chosen at random from the set $\{1,2,3 \ldots \ldots 3n\}$ then probability that x^2-y^2 is divisible by 3

(A) $\frac{5n-3}{3(3n-1)}$

(B) $\frac{5n-2}{3(3n-1)}$

(C) $\frac{n-2}{3n-1}$

(D) $\frac{1}{3n-1}$

Q.4

A bag contains a large number of white and black marbles in equal proportions. Two samples of 5 marbles are selected (with replacement) at random. The probability that the first sample contains exactly 1 black marble, and the second sample contains exactly 3 black marbles, is

(A) $\frac{25}{512}$

(B) $\frac{15}{32}$

(C) $\frac{15}{1024}$

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(D) $\frac{35}{256}$

Q.5

Suppose families always have one, two or three children, with probabilities $\frac{1}{4}, \frac{1}{2}$ and $\frac{1}{4}$ respectively. Assume everyone eventually gets married and has children, the probability of a couple having exactly four grandchildren is

(A) $\frac{27}{128}$

(B) $\frac{37}{128}$

(C) $\frac{25}{128}$

(D) $\frac{20}{128}$

Q.6

Let A and B be two events such that $P(A \cup B) \geq 3/4$ and $1/8 \leq P(A \cap B) \leq 3/8$

Statement $-1:\quad P(A)+P(B) \geq 7/8$

Statement $-2:\quad P(A)+P(B) \leq 11/8$

(A) Only I is true

(B) Both I, II are true

(C) Only II is true

(D) neither I nor II is true

Q.7

A fair coin is tossed repeatedly until two consecutive heads is obtained. The probability that two consecutive heads occur on the seventh and eights flips is equal to :

(A) $\frac{11}{256}$

(B) $\frac{15}{256}$

(C) $\frac{13}{256}$

(D) $\frac{17}{256}$

Q.8

An insurance company believes that people can be divided into two classes; those who are accident prone and those who are not. Their statistics show that an accident prone person will not have an accident in a yearly period with probability 0.4 whereas this probability is 0.2 for the other kind.Given that 30% of people are accident prone. The probability that a new policyholder will have an accident within a year of purchasing a policy is

(A) 0.74

(B) 0.28

(C) 0.34

(D) 0.66

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Q.9

If for the events A and B, $P(B) = P\left(\frac{B}{A}\right) = \frac{1}{4}$ and $P\left(\frac{A}{B}\right) = \frac{1}{2}$ then $-$

(A) A and B are independent

(B) $\mathrm{P}\left(\frac{\mathrm{B}'}{\mathrm{A}}\right) = \frac{3}{4}$

(C) $\mathrm{P}\left(\frac{\mathrm{A}'}{\mathrm{B}'}\right) = \frac{1}{2}$

(D) A and B are mutually exclusive

Q.10

If A and B are two events such

that $\mathrm{P}(\mathrm{A}) = 3/4$ and $\mathrm{P}(\mathrm{B}) = 5/8$, then $-$

(A) $\mathrm{P}(\mathrm{A} \cap \mathrm{B}) < 3/8$

(B) $P(A \cap B) \geq 3/8$

(C) $P(A \cup B) = 11/16$

(D) $P(A \cap B) < 11/16$

Q.11

A die is rolled three times. Let A denote the event of getting a number larger than the previous number each time, and B denote the event the number (in order) form an increasing arithmetic progression. Then -

(A) $P(A) > P(B)$

(B) $P(A \cap B) = \frac{1}{36}$

(C) $\mathrm{P}\left(\frac{\mathrm{B}}{\mathrm{A}}\right) = \frac{3}{10}$

(D) $P(A) = 3P(B)$

Q.12

Passage:

A contest is arranged in the following manner. A contestant P has to play cards games with 4 players $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D separately. P is declared to have won the contest if he/she wins 3 consecutive games out of the four games played, otherwise, P loses the contest. However, P can choose in which order to play the games, for example, the order of play can be ACBD or BDAC or DBCA and so on. Also given the following information.

Odds in favour of P winning the game with A is 5 to 6 .

Odds in favour of P winning the game with B is 2 to 5 .

Odds in favour of P winning the game with C is 4 to $7.$

and odds in favour of P winning the game with D is 3 to 2 .

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Question:

If P plays the games in the order ADBC, the probability of his/her winning the contest is -

(A) $\frac{410}{4235}$

(B) $\frac{474}{4235}$

(C) $\frac{344}{4235}$

(D) $\frac{500}{4235}$

Q.13

Passage:

A contest is arranged in the following manner. A contestant P has to play cards games with 4 players $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D separately. P is declared to have won the contest if he/she wins 3 consecutive games out of the four games played, otherwise, P loses the contest. However, P can choose in which order to play the games, for example, the order of play can be ACBD or BDAC or DBCA and so on. Also given the following information.

Odds in favour of P winning the game with A is 5 to 6 .

Odds in favour of P winning the game with B is 2 to 5 .

Odds in favour of P winning the game with C is 4 to $7.$

and odds in favour of P winning the game with D is 3 to 2 .

Question:

If P plays the games first with B and second with A, the probability of his/her winning the contest is

(A) $\frac{314}{847}$

(B) $\frac{210}{847}$

(C) $\frac{217}{847}$

(D) $\frac{226}{847}$

Q.14

Passage:

A contest is arranged in the following manner. A contestant P has to play cards games with 4 players $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D separately. P is declared to have won the contest if he/she wins 3 consecutive games out of the four games played, otherwise, P loses the contest. However, P can choose in which order to play the games, for example, the order of play can be ACBD or BDAC or DBCA and so on. Also given the following information.

Odds in favour of P winning the game with A is 5 to 6 .

Odds in favour of P winning the game with B is 2 to 5 .

Odds in favour of P winning the game with C is 4 to $7.$

and odds in favour of P winning the game with D is 3 to 2 .

Question:

If P plays the games first with D, then the probability of P winning the contest is $-$

(A) $\frac{228}{385}$

(B) $\frac{226}{847}$

(C) $\frac{344}{847}$

(D) $\frac{288}{847}$

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Q.15

Passage:
A boy throws stones at a target with equal chance of hitting or missing it in any given attempt. He will stop when he hits the target 5 times.

Question:
Probability that he stops after 10th trial is -

(A) $\frac{63}{1024}$

(B) $\frac{63}{512}$

(C) $\frac{61}{512}$

(D) $\frac{63}{2048}$

Q.16

Passage:
A boy throws stones at a target with equal chance of hitting or missing it in any given attempt. He will stop when he hits the target 5 times.

Question:
Probability that he requires a minimum of 10 trials is -

(A) $\frac{449}{512}$

(B) $\frac{63}{512}$

(C) $\frac{121}{128}$

(D) $\frac{7}{128}$

Q.17

Passage:
A boy throws stones at a target with equal chance of hitting or missing it in any given attempt. He will stop when he hits the target 5 times.

Question:
Probability that he requires at least 12 trials given that he requires more than 10 trials is -

(A) $\frac{231}{1024}$

(B) $\frac{281}{1024}$

(C) $\frac{281}{512}$

(D) $\frac{281}{386}$

Q.18

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Match the column -

	Column I	Column II
(A)	A pair of dice rolled together till a sum of 5 or 7 is obtained. The probability that 5 comes before 7 is	(p) 7/8
(B)	The probability that in a family of 3 children there will be at least one girl is	(q) 2/9
(C)	Two points are selected in a line AC of length 'a' so as to lie on the opposite sides of the middle point of AC. The probability that the distance between them is < a/3 is	(r) 1/2
(D)	A natural number is selected at random from the first 100 natural numbers. The probability that the number satisfied the inequality $x^2 - 17x \leq 60$ is	(s) 2/5 (t) 1/5

(A) (A) - q, (B) - r, (C) - p, (D) - t

(B) (A) - r, (B) - q, (C) - s, (D) - r

(C) (A) - s, (B) - p, (C) - q, (D) - t

(D) (A) - s, (B) - t, (C) - s, (D) - p

Q.19

Suppose $f(x) = x^3 + ax^2 + bx + c$, where a, b, c are chosen respectively by throwing a die three times. If the probability that $f(x)$ is an increasing function is P then find $\frac{9P}{4}$.

Q.20

A ball is drawn at random from a bag containing 3 balls and it is found to be black. If it is equally likely that it may contain 1 or 2 or 3 black balls, if probability that the ball drawn is the only black ball is P then find $12P$.

Answers & Solutions

Q.1 Answer:

$\frac{13}{60}$

Solution:

From set theory

$P(B) = P(A \cap B \cap C^c) + (A^c \cap B \cap C^c) + P(B \cap C)$

$\frac{4}{5} = \frac{1}{4} + \frac{1}{3} + P(B \cap C)$

$\Rightarrow P(B \cap C) = \frac{4}{5} - \frac{1}{4} - \frac{1}{3} = \frac{48-15-20}{60}$

$= \frac{13}{60}$

Q.2 Answer:

$\frac{1}{n}\left[1 - (1-p)^n\right]$

Solution:

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(1) Let Ei denote the event that Ai dies in a year, then $\mathrm{P}\left(\mathrm{E}_{\mathrm{i}}\right)=\mathrm{p}$ **and** $\mathrm{P}\left(\mathrm{E}_{\mathrm{i}}^{\prime}\right)=1-\mathrm{p}$ **for** $\mathrm{I}=1,$ $2, \ldots .\mathrm{nP}$ **(none of** $\mathrm{A}_1, \mathrm{A}_2, \ldots . \mathrm{A}_3$ **dies in a year)**

$=P\left(E_1^{\prime} \cap E_2^{\prime} \cap E_{\mathrm{n}}^{\prime}\right)=P\left(E_1^{\prime}\right) P\left(E_2^{\prime}\right) \ldots . P\left(E_n^{\prime}\right)=(1-p)^{\mathrm{n}}$

Because $\mathrm{E}_1, \mathrm{E}_2, \ldots \ldots \mathrm{E}_{\mathrm{n}}$ **are independent.**

Let E **denote the event that at least one of** $\mathrm{A}_1, \mathrm{A}_2, \ldots \ldots \mathrm{A}_{\mathrm{n}}$ **dies in a year, then** $\mathrm{P}(\mathrm{E})=1-$ $\mathrm{P}\left(\mathrm{E}_1^{\prime} \cap \mathrm{E}_2^{\prime} \cap \ldots \cap \mathrm{E}_{\mathrm{n}}^{\prime}\right)=1-(1-\mathrm{p})^{\mathrm{n}}$

Let F **denote the event that** A_1 **is the first to die, then** $\mathrm{P}(\mathrm{F} / \mathrm{E})=\frac{1}{\mathrm{n}}$**.Also,** $\mathrm{P}(\mathrm{F})=\mathrm{P}(\mathrm{E}) \cdot \mathrm{P}(\mathrm{F} / \mathrm{E})=$ $\frac{1}{n}\left[1-(1-p)^n\right]$

Q.3 Answer:

$\frac{5n-3}{3(3n-1)}$

Solution:

$$P(E)=\frac{3\left(n_{C_2}\right)+\left(n_{C_1}\right)^2}{(3n)_{C_2}}=\frac{5n-3}{3(3n-1)}$$

Q.4 Answer:

$\frac{25}{512}$

Solution:

Let the number of marble be $2n$ **(where** n **is large)**

$$\text{Required probability } =\lim_{\mathrm{n}\to\infty}\frac{\mathrm{n}\times{}^{\mathrm{n}}\mathrm{C}_4}{{}^{2n}C_5}\times\frac{{}^{\mathrm{n}}\mathrm{C}_3\times{}^{\mathrm{n}}\mathrm{C}_2}{{}^{2n}C_5}$$

$$=\lim_{\mathrm{n}\to\infty}\frac{\mathrm{n}\times\mathrm{n}(\mathrm{n}-1)(\mathrm{n}-2)(\mathrm{n}-3)}{4!}\times\frac{\mathrm{n}(\mathrm{n}-1)(\mathrm{n}-2)}{3!}\times\frac{\mathrm{n}(\mathrm{n}-1)}{2!}\times\frac{(5)^2((2n-5)!)^2}{(2n!)^2}$$

$$=\lim_{n\to\infty}\frac{n^4(n-1)^3(n-2)^2(n-3)((2n-5)!)^2\times5\times5\times4\times3!}{3!2!(2n!)^2}$$

$$=\lim_{n\to\infty}\frac{50\cdot n^4(n-1)^3(n-2)^2(n-3)}{(2n(2n-1)(2n-2)(2n-3)(2n-4))^2}=\frac{50}{1024}=\frac{25}{512}.$$

Q.5 Answer:

$\frac{27}{128}$

Solution:

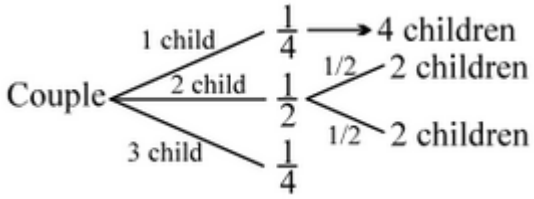
A : exactly one child

B **: exactly two children**

C **: exactly 3 childre**

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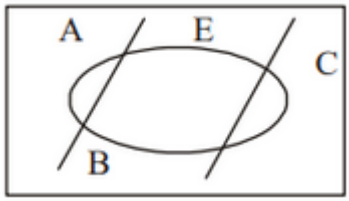
$P(A) = \frac{1}{4}; P(B) = \frac{1}{2}; P(C) = \frac{1}{4}$



E : couple has exactly 4 grandchildren

$$P(E) = P(A) \cdot P(E/A) + P(B) \cdot P(E/B) + P(C) \cdot P(E/C)$$

$$= \underbrace{\frac{1}{4} \cdot 0}_{\substack{\text{onechild} \\ \text{and have} \\ \text{morethan} \\ \text{3children}}} + \frac{1}{2}\left[\underbrace{\left(\frac{1}{2}\right)^2}_{2/2} + \underbrace{\frac{1}{4} \cdot \frac{1}{4} \cdot 2}_{(1,3) \text{ or } (3,1)}\right] + \frac{1}{4}\left[3\underbrace{\left(\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}\right)}_{1}\right]$$



$= \frac{1}{8} + \frac{1}{16} + \frac{3}{128} = \frac{27}{128}$ **Ans.**

Similarly, 2 / 2 denotes each child having two children; ' 0 ' indicated that the child can have a maximum of 3 children

$2 \cdot \frac{1}{4} \cdot \frac{1}{4}$ **denotes each child having 1 and 3 or 3 and 1 children** $= \frac{16}{128} + \frac{8}{128} + \frac{3}{128} = \frac{27}{128}$ **Ans.**

Q.6 Answer:

Both I, II are true

Solution:

$$P(A \cup B) \geq \frac{3}{4} \Rightarrow P(A) + P(B) \geq \frac{3}{4} + P(A \cap B)$$
$$\geq \frac{3}{4} + \frac{1}{8} \geq \frac{7}{8}$$
$$P(A \cup B) \leq 1 \Rightarrow P(A) + P(B) \leq 1 + \frac{3}{8}$$
$$\leq \frac{11}{8}$$

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Q.7 Answer:

$\frac{13}{256}$

Solution:

XXXXXTHH

Now at 6th place tail must be present. Possible cases in first 5 places.

Case-I: T TTTT All tails=1 ways;

Case-II: 4T and H i.e., T TTT H $\Rightarrow \frac{5!}{4!} = 5$ **ways**

Case-III: 3 T and 2H but not consecutive heads.

T TT (4 gaps) and place 2 heads in gaps $\Rightarrow {}^4C_2 = 6$ **ways**

Case-IV: 2 T and 3H but not consecutive heads.

T T (3 gaps) and places 3 heads in gaps $\Rightarrow {}^3C_3 = 1$ **ways**

$\Rightarrow$ **Total 13 cases and in every case probability is** $\frac{1}{256}$.

$\therefore$ **Required probability** $= \frac{13}{256}$

Q.8 Answer:

0.74

Solution:

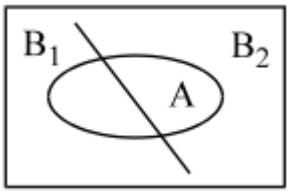
A : policyholder has an accident within a year.

B_1 **: policyholder is accident prone.**

B_2 **: policyholder is not accident prone.**

Given $P(B_1) = \frac{3}{10}; P(B_2) = \frac{7}{10};$

$P(A/B_1) = 0.6; \quad P(A/B_2) = 0.8$



$$P(A) = P(A \cap B_1) + P(A \cap B_2) = P(B_1)\,P(A/B_1) + P(B_2)\,P(A/B_2)$$
$$= \frac{3}{10} \times \frac{6}{10} + \frac{7}{10} \times \frac{8}{10} = \frac{74}{100} = \frac{37}{50} = 0.74 \text{ Ans.}$$

Q.9 Answer:

A and B are independent

Solution:

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$P(B) = \left(\frac{B}{A}\right) \Rightarrow A$ **and** B **are independent.**

Also $\mathrm{P}\left(\frac{\mathrm{B}'}{\mathrm{A}}\right) = \mathrm{P}\left(\mathrm{B}'\right) = 1 - \mathrm{P}(\mathrm{B}) = 1 - \frac{1}{4} = \frac{3}{4}$

Since A **and** B **are independent,** $\mathrm{P}\left(\frac{\mathrm{A}}{\mathrm{B}}\right) = \frac{1}{2} \Rightarrow \mathrm{P}(\mathrm{A}) = \frac{1}{2}$

$\therefore \quad \mathrm{P}\left(\mathrm{A}'\right) = \frac{1}{2}$**. Also** A **and** B **are independent** $\Rightarrow \mathrm{A}'$ **and** B' **are also independent**

$\therefore \quad \mathrm{P}\left(\frac{\mathrm{A}'}{\mathrm{B}'}\right) = \mathrm{P}\left(\mathrm{A}'\right) = \frac{1}{2}$

D is wrong since independent is not related to mutually exclusive.

Q.10 Answer:

$P(A \cap B) \geq 3/8$

Solution:

$\mathrm{P}(\mathrm{A} \cap \mathrm{B}) = \mathrm{P}(\mathrm{A}) + \mathrm{P}(\mathrm{B}) - \mathrm{P}(\mathrm{A} \cup \mathrm{B}) \geq \frac{3}{4} + \frac{5}{8} - 1 \quad [\because \mathrm{P}(\mathrm{A} \cup \mathrm{B}) \leq 1] = \frac{3}{8}$

Also, $\mathrm{P}(\mathrm{A} \cap \mathrm{B}) \leq \mathrm{P}(\mathrm{B}) \Rightarrow \mathrm{P}(\mathrm{A} \cap \mathrm{B}) \leq \frac{5}{8} < \frac{11}{16}$

Q.11 Answer:

$P(A) > P(B)$

Solution:

The number of exhaustive cases $= 6 \times 6 \times 6 = 216$

For the event of A **the** 2nd **number should be form 2 to 5.**

If the 2nd **number is** i**, then the first number is** $i - 1$ **and the** 3rd **number of** $6 - i$**.**

$\therefore$ **The number of favourable cases for the event**

$A = \sum_{i=2}^{5}(i-1)(6-i) = 20$

$P(A) = \frac{20}{216} = \frac{5}{54}$

Sample space for the event B

$\{(1,2,3)(2,3,4)(3,4,5)(4,5,6)(1,3,5)(2,4,6)\}$

$\mathrm{P}(\mathrm{B}) = \frac{6}{216} = \frac{1}{36}$

$\mathrm{B} \subseteq \mathrm{A} \Rightarrow \mathrm{P}(\mathrm{B}) = \mathrm{P}(\mathrm{A} \cap \mathrm{B}) = \frac{1}{36}$

(A) is correct

(B) is correct

$P\left(\frac{B}{A}\right) = \frac{\frac{1}{36}}{\frac{5}{54}} = \frac{3}{10}$

Evidently (D) is not correct.

Q.12 Answer:

$\frac{474}{4235}$

Solution:

If $P(A), P(B), P(C), P(D)$ **denote the probabilities of** P **winning the games with the players** A**,** B, C, D **then** $P(A) = \frac{5}{11}, P(B) = \frac{2}{7}, P(C) = \frac{4}{11}, P(D) = \frac{3}{5}$

(i) Since the order of play in ADBC**,** P **will win the contest if (i)** P **win with** $\mathrm{A}, \mathrm{D}, \mathrm{B}$ **or (ii)** P **loses with**

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A **but wins with** D, B, C.

Required probability $= \mathrm{P(ADB)} + \mathrm{P\,(A'DBC)} = \frac{5}{11} \times \frac{3}{8} \times \frac{2}{7} + \frac{6}{11} \times \frac{3}{5} \times \frac{2}{7} \times \frac{4}{11} = \frac{474}{4235}$

(ii) BACD, BADC are two possibilities

Required probability $= \mathrm{P(BAC)} + \mathrm{P\,(B'ACD)} + \mathrm{P(BAD)} + \mathrm{P\,(B'ADC)}$

$= \frac{2}{7} \times \frac{5}{11} \times \frac{4}{11} + \frac{5}{7} \times \frac{5}{11} \times \frac{4}{11} \times \frac{3}{8} + \frac{2}{7} \times \frac{5}{11} \times \frac{3}{8} + \frac{5}{7} \times \frac{5}{11} \times \frac{3}{8} \times \frac{4}{11} = \frac{1130}{4235} = \frac{226}{847}$

(iii) The various possibilities are DABC, DACB, DBCA, DBAC, DCAB, DCBA.

$\mathrm{DABC} : \mathrm{P(DAB)} + \mathrm{P\,(D'ABC)} = \frac{3}{5} \times \frac{5}{11} \times \frac{2}{7} + \frac{2}{5} \times \frac{5}{11} \times \frac{2}{7} \times \frac{4}{11} = \frac{410}{4235}$

$\mathrm{DACB} : \mathrm{P(DAC)} + \mathrm{P\,(D'ACB)} = \frac{3}{5} \times \frac{5}{11} \times \frac{4}{11} + \frac{80}{4235} = \frac{500}{4235}$

$\mathrm{DBCA} : \mathrm{P(DBC)} + \mathrm{P\,(D'BCA)} = \frac{3}{5} \times \frac{2}{7} \times \frac{4}{11} + \frac{80}{4235} = \frac{344}{4235}$

$\mathrm{DBAC} : \mathrm{P(DBA)} + \mathrm{P\,(D'BAC)} = \frac{3}{5} \times \frac{2}{7} \times \frac{5}{11} + \frac{80}{4235} = \frac{410}{4235}$

$\mathrm{DCAB} : \mathrm{P(DCA)} + \mathrm{P\,(D'CAB)} = \frac{3}{5} \times \frac{4}{11} \times \frac{5}{11} + \frac{80}{4235} = \frac{500}{4235}$

$\mathrm{DCBA} : \mathrm{P(DCB)} + \mathrm{P\,(D'CBA)} = \frac{3}{5} \times \frac{4}{11} \times \frac{2}{7} + \frac{80}{4235} = \frac{344}{4235}$

Required probability

= Sum of the probabilities corresponding to the above 6 mutually exclusive cases

$= \frac{2508}{4235} = \frac{228}{385}$

Q.13 Answer:

$\frac{226}{847}$

Solution:

If $P(A), P(B), P(C), P(D)$ **denote the probabilities of** P **winning the games with the players** A, B, C, D **then** $P(A) = \frac{5}{11}, P(B) = \frac{2}{7}, P(C) = \frac{4}{11}, P(D) = \frac{3}{5}$

(i) Since the order of play in ADBC, P **will win the contest if (i)** P **win with** $\mathrm{A}, \mathrm{D}, \mathrm{B}$ **or (ii)** P **loses with** A **but wins with** D, B, C.

Required probability $= \mathrm{P(ADB)} + \mathrm{P\,(A'DBC)} = \frac{5}{11} \times \frac{3}{8} \times \frac{2}{7} + \frac{6}{11} \times \frac{3}{5} \times \frac{2}{7} \times \frac{4}{11} = \frac{474}{4235}$

(ii) BACD, BADC are two possibilities

Required probability $= \mathrm{P(BAC)} + \mathrm{P\,(B'ACD)} + \mathrm{P(BAD)} + \mathrm{P\,(B'ADC)}$

$= \frac{2}{7} \times \frac{5}{11} \times \frac{4}{11} + \frac{5}{7} \times \frac{5}{11} \times \frac{4}{11} \times \frac{3}{8} + \frac{2}{7} \times \frac{5}{11} \times \frac{3}{8} + \frac{5}{7} \times \frac{5}{11} \times \frac{3}{8} \times \frac{4}{11} = \frac{1130}{4235} = \frac{226}{847}$

(iii) The various possibilities are DABC, DACB, DBCA, DBAC, DCAB, DCBA.

$\mathrm{DABC} : \mathrm{P(DAB)} + \mathrm{P\,(D'ABC)} = \frac{3}{5} \times \frac{5}{11} \times \frac{2}{7} + \frac{2}{5} \times \frac{5}{11} \times \frac{2}{7} \times \frac{4}{11} = \frac{410}{4235}$

$\mathrm{DACB} : \mathrm{P(DAC)} + \mathrm{P\,(D'ACB)} = \frac{3}{5} \times \frac{5}{11} \times \frac{4}{11} + \frac{80}{4235} = \frac{500}{4235}$

$\mathrm{DBCA} : \mathrm{P(DBC)} + \mathrm{P\,(D'BCA)} = \frac{3}{5} \times \frac{2}{7} \times \frac{4}{11} + \frac{80}{4235} = \frac{344}{4235}$

$\mathrm{DBAC} : \mathrm{P(DBA)} + \mathrm{P\,(D'BAC)} = \frac{3}{5} \times \frac{2}{7} \times \frac{5}{11} + \frac{80}{4235} = \frac{410}{4235}$

$\mathrm{DCAB} : \mathrm{P(DCA)} + \mathrm{P\,(D'CAB)} = \frac{3}{5} \times \frac{4}{11} \times \frac{5}{11} + \frac{80}{4235} = \frac{500}{4235}$

$\mathrm{DCBA} : \mathrm{P(DCB)} + \mathrm{P\,(D'CBA)} = \frac{3}{5} \times \frac{4}{11} \times \frac{2}{7} + \frac{80}{4235} = \frac{344}{4235}$

Required probability

= Sum of the probabilities corresponding to the above 6 mutually exclusive cases

$= \frac{2508}{4235} = \frac{228}{385}$

Q.14 Answer:

$\frac{228}{385}$

Solution:

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If $P(A), P(B), P(C), P(D)$ denote the probabilities of P winning the games with the players A, B, C, D then $P(A)=\frac{5}{11}, P(B)=\frac{2}{7}, P(C)=\frac{4}{11}, P(D)=\frac{3}{5}$

(i) Since the order of play in ADBC, P will win the contest if (i) P win with A, D, B or

(ii) P loses with A but wins with D, B, C.

Required probability $=\mathrm{P}(\mathrm{ADB})+\mathrm{P}\left(\mathrm{A}'\mathrm{DBC}\right)=\frac{5}{11}\times\frac{3}{8}\times\frac{2}{7}+\frac{6}{11}\times\frac{3}{5}\times\frac{2}{7}\times\frac{4}{11}=\frac{474}{4235}$

(ii) BACD, BADC are two possibilities

Required probability $=\mathrm{P}(\mathrm{BAC})+\mathrm{P}\left(\mathrm{B}'\mathrm{ACD}\right)+\mathrm{P}(\mathrm{BAD})+\mathrm{P}\left(\mathrm{B}'\mathrm{ADC}\right)$

$=\frac{2}{7}\times\frac{5}{11}\times\frac{4}{11}+\frac{5}{7}\times\frac{5}{11}\times\frac{4}{11}\times\frac{3}{8}+\frac{2}{7}\times\frac{5}{11}\times\frac{3}{8}+\frac{5}{7}\times\frac{5}{11}\times\frac{3}{8}\times\frac{4}{11}=\frac{1130}{4235}=\frac{226}{847}$

(iii) The various possibilities are DABC, DACB, DBCA, DBAC, DCAB, DCBA.

$\mathrm{DABC}:\mathrm{P}(\mathrm{DAB})+\mathrm{P}\left(\mathrm{D}'\mathrm{ABC}\right)=\frac{3}{5}\times\frac{5}{11}\times\frac{2}{7}+\frac{2}{5}\times\frac{5}{11}\times\frac{2}{7}\times\frac{4}{11}=\frac{410}{4235}$

$\mathrm{DACB}:\mathrm{P}(\mathrm{DAC})+\mathrm{P}\left(\mathrm{D}'\mathrm{ACB}\right)=\frac{3}{5}\times\frac{5}{11}\times\frac{4}{11}+\frac{80}{4235}=\frac{500}{4235}$

$\mathrm{DBCA}:\mathrm{P}(\mathrm{DBC})+\mathrm{P}\left(\mathrm{D}'\mathrm{BCA}\right)=\frac{3}{5}\times\frac{2}{7}\times\frac{4}{11}+\frac{80}{4235}=\frac{344}{4235}$

$\mathrm{DBAC}:\mathrm{P}(\mathrm{DBA})+\mathrm{P}\left(\mathrm{D}'\mathrm{BAC}\right)=\frac{3}{5}\times\frac{2}{7}\times\frac{5}{11}+\frac{80}{4235}=\frac{410}{4235}$

$\mathrm{DCAB}:\mathrm{P}(\mathrm{DCA})+\mathrm{P}\left(\mathrm{D}'\mathrm{CAB}\right)=\frac{3}{5}\times\frac{4}{11}\times\frac{5}{11}+\frac{80}{4235}=\frac{500}{4235}$

$\mathrm{DCBA}:\mathrm{P}(\mathrm{DCB})+\mathrm{P}\left(\mathrm{D}'\mathrm{CBA}\right)=\frac{3}{5}\times\frac{4}{11}\times\frac{2}{7}+\frac{80}{4235}=\frac{344}{4235}$

Required probability

$=$ Sum of the probabilities corresponding to the above 6 mutually exclusive cases

$=\frac{2508}{4235}=\frac{228}{385}$

Q.15 Answer:

$\frac{63}{512}$

Solution:

(i) Let us denote a hit by H and a miss by M.

The boy wants 5 hits.

Let X denote the number of attempts required to stop the experiment.

Then X can take any value $\mathrm{n}=5,6,7,\ldots\ldots\ldots$

To stop after n^{th} trial the boy should have had 4 successes in $(n-1)$ trials and the nth trial should be a success.

$\therefore \mathrm{p}[\mathrm{x}=\mathrm{n}]={}^{\mathrm{n}-1}\mathrm{C}_4\mathrm{p}^4\mathrm{q}^{\mathrm{n}-1-4}\times\mathrm{p}={}^{\mathrm{n}-1}\mathrm{C}_4\mathrm{p}^5\mathrm{q}^{\mathrm{n}-5},\mathrm{n}=5,6,7,\ldots\ldots\ldots$

Where $\mathrm{p}=$ probability of success in a trial and $\mathrm{q}=$ probability of failure.

Here $\mathrm{p}=\mathrm{q}=\frac{1}{2}$ (equal change of a hit or miss)

$\mathrm{n}=10$

$$\therefore \mathrm{p}[\mathrm{x}=10]={}^{9}\mathrm{C}_4\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^{10-5}=\frac{1}{2^{10}}\times{}^{9}\mathrm{C}_4=\frac{63}{512}$$

(ii) $\mathrm{p}(\mathrm{x}\geq 10)=1-\mathrm{p}(\mathrm{x}\leq 9)=1-(\mathrm{p}(\mathrm{x}=5)+\mathrm{p}(\mathrm{x}=6)+\mathrm{p}(\mathrm{x}=7)+\mathrm{p}(\mathrm{x}=8)+\mathrm{p}(\mathrm{x}=9))$

$=1-(1/2)^{10}\left[{}^{4}\mathrm{C}_4+{}^{5}\mathrm{C}_4+{}^{6}\mathrm{C}_4+{}^{7}\mathrm{C}_4+{}^{8}\mathrm{C}_4\right]=\frac{449}{512}$

(iii) Required probability $=\mathrm{p}[\mathrm{x}\geq 12/\mathrm{x}>10]$

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$= \frac{p[x \geq 12 \cap x > 10]}{p[x > 10]} = \frac{p[x \geq 12]}{p[x > 10]} \ldots\ldots\ldots\ldots (1)$

$p[x > 10] = p(x \geq 10) - p(x = 10)$

$= \frac{449}{512} - \frac{63}{512} = \frac{193}{256} \ldots\ldots\ldots\ldots\ldots (2)$

$p(x \geq 12) = p(x > 10) - p(x = 11)$

$= \frac{193}{256} - \frac{1}{2^{10}} 10C_4 = \frac{193}{256} - \frac{105}{512} = \frac{281}{512} \ldots\ldots\ldots (3)$

Using (3) and (2) in (1) gives the required probability $= \frac{281/512}{193/256} = \frac{281}{386}$

Q.16 Answer:

$\frac{449}{512}$

Solution:

(i) Let us denote a hit by H **and a miss by** M**. The boy wants 5 hits.**

Let X **denote the number of attempts required to stop the experiment. Then** X **can take any value** $n = 5, 6, 7, \ldots\ldots\ldots$

To stop after n^{th} **trial the boy should have had 4 successes in** $(n-1)$ **trials and the** n^{th} **trial should be a success.**

$\therefore p[x = n] = {}^{n-1}C_4 p^4 q^{n-1-4} \times p = {}^{n-1}C_4 p^5 q^{n-5}, n = 5, 6, 7, \ldots\ldots\ldots$

Where $p =$ **probability of success in a trial and** $q =$ **probability of failure.**

Here $p = q = \frac{1}{2}$ **(equal change of a hit or miss)**

$n = 10$

$$\therefore p[x = 10] = {}^9C_4 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{10-5} = \frac{1}{2^{10}} \times {}^9C_4 = \frac{63}{512}$$

(ii) $p(x \geq 10) = 1 - p(x \leq 9) = 1 - (p(x = 5) + p(x = 6) + p(x = 7) + p(x = 8) + p(x = 9))$

$= 1 - (1/2)^{10} \left[{}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4 + {}^8C_4\right] = \frac{449}{512}$

(iii) Required probability $= p[x \geq 12 / x > 10]$

$= \frac{p[x \geq 12 \cap x > 10]}{p[x > 10]} = \frac{p[x \geq 12]}{p[x > 10]} \ldots\ldots\ldots\ldots (1)$

$p[x > 10] = p(x \geq 10) - p(x = 10)$

$= \frac{449}{512} - \frac{63}{512} = \frac{193}{256} \ldots\ldots\ldots\ldots\ldots (2)$

$p(x \geq 12) = p(x > 10) - p(x = 11)$

$= \frac{193}{256} - \frac{1}{2^{10}} 10C_4 = \frac{193}{256} - \frac{105}{512} = \frac{281}{512} \ldots\ldots\ldots (3)$

Using (3) and (2) in (1) gives the required probability $= \frac{281/512}{193/256} = \frac{281}{386}$

Q.17 Answer:

$\frac{281}{386}$

Solution:

(i) Let us denote a hit by H **and a miss by** M**. The boy wants 5 hits.**

Let X **denote the number of attempts required to stop the experiment. Then** X **can take any value** $n = 5, 6, 7, \ldots\ldots\ldots$

To stop after n^{th} **trial the boy should have had 4 successes in** $(n-1)$ **trials and the** n^{th} **trial should be a success.**

$\therefore p[x = n] = {}^{n-1}C_4 p^4 q^{n-1-4} \times p = {}^{n-1}C_4 p^5 q^{n-5}, n = 5, 6, 7, \ldots\ldots\ldots$

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Where p = probability of success in a trial and q = probability of failure.

Here $p = q = \frac{1}{2}$ **(equal change of a hit or miss)**

$n = 10$

$$\therefore p[x = 10] = {}^9C_4 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{10-5} = \frac{1}{2^{10}} \times {}^9C_4 = \frac{63}{512}$$

(ii) $p(x \geq 10) = 1 - p(x \leq 9) = 1 - (p(x = 5) + p(x = 6) + p(x = 7) + p(x = 8) + p(x = 9))$

$= 1 - (1/2)^{10} \left[{}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4 + {}^8C_4\right] = \frac{449}{512}$

(iii) Required probability $= p[x \geq 12/x > 10]$

$= \frac{p[x \geq 12 \cap x > 10]}{p[x > 10]} = \frac{p[x \geq 12]}{p[x > 10]}$ .....(1)

$p[x > 10] = p(x \geq 10) - p(x = 10)$

$= \frac{449}{512} - \frac{63}{512} = \frac{193}{256}$ .....(2)

$p(x \geq 12) = p(x > 10) - p(x = 11)$

$= \frac{193}{256} - \frac{1}{2^{10}} 10C_4 = \frac{193}{256} - \frac{105}{512} = \frac{281}{512}$ .....(3)

Using (3) and (2) in (1) gives the required probability $= \frac{281/512}{193/256} = \frac{281}{386}$

Q.18 Answer:

(A) - s, (B) - p, (C) - q, (D) - t

Solution:

$(A) - s, (B) - p, (C) - q, (D) - t$

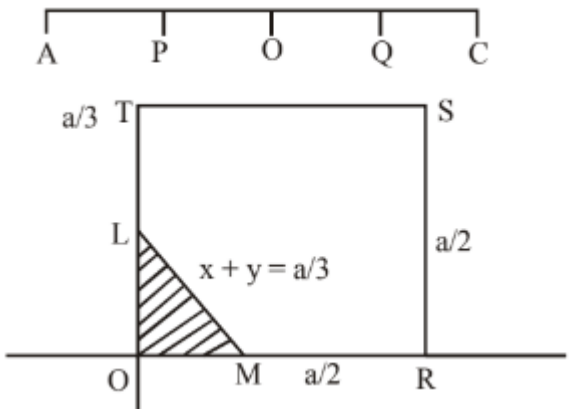
(A) The probability of getting $5 = \frac{4}{36} = \frac{1}{9} \quad \{(1,4),(2,3),(3,2),(4,1)\}$ **The probability of not getting either 5 or 7 while throwing 2 die** $= 1 - \frac{1}{9} - \frac{1}{6} = \frac{13}{18}$

$\therefore$ **The required probability** $= \frac{1}{9} + \frac{13}{18} \cdot \frac{1}{9} + \left(\frac{13}{18}\right)^2 \cdot \frac{1}{9} + \ldots\ldots$ **to** $\infty = \frac{1}{9}\left(\frac{1}{1-\frac{13}{18}}\right) = \frac{2}{5}$

(B) The probability that there will be at least one girl $= 1 - P($ **no girl** $) = 1 - \frac{1}{2^3} = \frac{7}{8}$

(C) Let $MP = x, \quad MQ = y$

$\therefore \quad x + y < \frac{a}{3}$ **.....(1)**



Also, $0 \leq x \leq \frac{a}{3}$ **.....(2)**

$0 \leq y \leq \frac{a}{3}$ **.....(3)**

Conditions (1), (2), (3) hold good for favourable cases.

For all possibilities $0 \leq x \leq \frac{a}{2}, 0 \leq y \leq \frac{a}{2}$

$\therefore$ **The required probability** $= \frac{\text{area of } \Delta \text{ OLM}}{\text{area of the square ORST}} = \frac{\frac{1}{2}\frac{a}{3}\frac{a}{3}}{\frac{a^2}{4}} = \frac{2}{9}$

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(D) $x^2 - 17x \leq 60 \Rightarrow x^2 - 17x - 60 \leq 0 \Rightarrow (x-20)(x+3) \leq 0$

$\Rightarrow 1 \leq x \leq 20$

$\Rightarrow$ **The required probability** $= \frac{20}{100} = \frac{1}{5}$

Q.19 Answer:

—

Solution:

$f'(x) = 3x^2 + 2ax + b$

$f(x)$ is increasing

$\therefore \quad f'(x) \geq 0 \quad x,$ where equally holds for discrete points.

$\therefore \quad D = (2a)^2 - 4 \times 3 \times b \leq 0 \Rightarrow a^2 - 3b \leq 0$

This is true for exactly 16 ordered pairs

$\therefore \frac{16}{36} = \frac{4}{9}.$

Q.20 Answer:

—

Solution:

Let A **be the event of drawing a black ball**

$$P\left(\frac{E_{iB}}{A}\right) = \frac{P\left(\frac{A}{E_{iB}}\right) \cdot P(E_{iB})}{\sum_{i=1}^{3} P\left(\frac{A}{E_{iB}}\right) \cdot P(E_{iB})} = \frac{1}{\frac{1}{3}[{}^1C_1 + {}^2C_1 + {}^3C_1]} = \frac{1}{6}$$

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